claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-ad97df50889e390fd.jsonl`
- SHA-256: `c128f2ee5a742b6d594a6ac5bf617e5515f2055e0f49baa9dae0f550d2922152`
- Source modified: `2026-06-19T17:19:25+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_4e32c8b8-725`
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`

Transcript

1. user (2026-06-19T17:18:10.923Z)

```
You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs
can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness
sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:
- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED.
"Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but
still-intended design/plan (owner-gated future work is LIVE, not terminal), a
reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template
(PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter
effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) ·
checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch
records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a
stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md,
docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an
archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag
DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere
(e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before
recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level
docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the
contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): ROORA_LABELING_GUIDELINES.md, TRACKNET_WSL_SETUP.md,
VIDEO_RECORDING_GUIDELINES.md, OWNED_MODEL_IMPLEMENTATION_PLAN.md,
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md

For EACH doc:
1. Read the status header + TL;DR/intro + (if present) the §7/§8 progress tracker + definition-
of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle
state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started
DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-
superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the
doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a
FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a
stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the
other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it
```

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.
5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.261Z)

I'll audit each doc in my batch. Let me start by reading them and gathering cross-reference evidence in parallel.

3. user (2026-06-19T17:18:16.721Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end'
from the remote, but no such ref was fetched.
---ARCHIVES---
CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
COMMERCIAL_READINESS_REVIEW.md
DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
MONETIZATION_AUDIT.md
README.md
TEST_RUN_SUMMARY.md
checkpoints
decisions
past_projects
quality-iterations
research

4. user (2026-06-19T17:18:16.741Z)

1 # TrackNet / WASB on WSL2 ? setup & usage
2
3 Self-hosted shuttle-trajectory detectors run inside WSL2 (Ubuntu) and are bridged
4 to the Windows indexer via a subprocess adapter. See
`docs/archives/decisions/DECISIONS.md` (ADR-001)
5 for the why.
6
7 ## What's installed (WSL2 / Ubuntu 24.04)
8
9 | Thing | Location (WSL) |
10 |-------|----------------|
11 | Miniconda | `~/miniconda3` |
12 | TrackNetV4 repo | `~/models/TrackNetV4` (https://github.com/TrackNetV4/TrackNetV4) |
13 | TrackNetV4 conda env | `TrackNetV4` (Python 3.9, TensorFlow 2.17 + bundled CUDA 12.3)
|
14 | WASB repo | `~/models/WASB-SBDT` (https://github.com/nttcom/WASB-SBDT) |
15 | WASB conda env | `wasb` (Python 3.8, torch 1.11.0+cu113) ? verified
`torch.cuda.is_available()==True` |
16 | Staging dir for videos | `~/clips` (avoids slow `/mnt/c` per-frame reads) |
17
18 GPU: RTX 2070 Super visible via `nvidia-smi` inside WSL (CUDA-on-WSL working).
19
20 Activate an env with:
21 ```bash
22 source ~/miniconda3/etc/profile.d/conda.sh
23 conda activate TrackNetV4   # or: conda activate wasb
24 ```
25
26 ## ?? Two upstream issues you must resolve before TrackNetV4 inference
27
28 1. **No pretrained weights are hosted.** The "Download" links in

   `~/models/TrackNetV4/docs/RESULT.md` are placeholders (`(#)`). You will obtain
   weights by **training** (`src/train.py`) on your annotated data ? which is the
   plan. Until you have a `.h5`/`.keras` weights file, the segmenter's healthcheck
   passes but `run_predict` will log an error that `weights_path` is unset.

2. **`predict.py` has a custom-objects bug.** `src/predict.py` references layers in
   `custom_objects` that it never imports (`MotionIncorporationLayerV1`,
   `MotionIncorporationLayerV2`, `CombineOutputs`, `MotionFramesInput`) ? it will
   raise `NameError` on `load_model`. The imported layers are
   `MotionPromptLayer, FusionLayerTypeA, FusionLayerTypeB`. Patch the
   `custom_objects` dict in `main()` to match the layers your trained model
   actually uses, e.g.:
   ```python
   model = load_model(model_weights_path, custom_objects={
       'MotionPromptLayer': MotionPromptLayer,
       'FusionLayerTypeA': FusionLayerTypeA,   # if you trained TrackNetV4_TypeA
       'FusionLayerTypeB': FusionLayerTypeB,   # if you trained TrackNetV4_TypeB
       'custom_loss': custom_loss,
   })
   ```

Wiring it into the indexer

The segmenter is `tracknet_hybrid` (`backend/pipeline/segmenters/tracknet_hybrid.py`), backed by
`backend/pipeline/detectors/tracknet_runner.py`. Configure it in `config.json` under
`indexing.tracknet`:

```jsonc
"tracknet": {
  "wsl_distro": "Ubuntu",
  "repo_dir": "~/models/TrackNetV4",
  "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
```

5. user (2026-06-19T17:18:16.742Z)

Roora Annotation Guidelines ? Rally Golden Set

Version: v8 (2026-06-13). Supersedes the older Drive copies (v3/v5/v6/v7) **and** the internal
"DRAFT v2" label ? those numbers never matched each other; **v8 is the single label to use with
Roora going forward.** Content is otherwise the finalized 2026-06-12 guideline plus the
fusion-driven additions marked **`? ADDED (fusion)`** below.

> **?? Canonical home / provenance.** **This in-repo file is the sole canonical copy of the guideline**
> ? it lives here because the multi-signal-fusion features that motivate the new fields (court
> calibration, singles/doubles) originate in *this* repo (see
> MULTI_SIGNAL_FUSION_PLAN.md and
> GOLDEN_SET_VENDORING.md, ADR-006). The annotation deliverable still imports
> into the sibling **`sports-data-collector`** repo via `python -m src.cli.curate import-local`; if a
> mirror is ever staged there for the collector's intake docs, it must be regenerated from this file
> (and the `? ADDED (fusion)` callouts are exactly the deltas to port).
>
> **`TODO(avidullu)` markers are preserved** ? they are owner-only items (Drive links, intake
> manifest, NDA counsel review) and must be filled before sending to Roora. Search for `TODO(avidullu)`.

Brief (vendor spec) + Illustrated Guide (rater how-to), in one doc.

PART A ? Roora Annotation Brief (vendor-facing spec)

Project: racket-sport rally segmentation golden / regression dataset.
Sports in scope: Badminton, Pickleball, Table Tennis (each in singles AND doubles).

This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that imports directly via `python -m src.cli.curate import-local`.

0. What we need in one sentence

For each provided match video, produce ONE CSV file listing EVERY rally with its start time, end time, number of shots, and how it ended ? EXCLUDING all dead time.

1. Deliverable format (canonical)

One CSV per video (UTF-8), named exactly `<video_id>.csv` (we supply the `video_id`). One row per rally, chronological, header row required.

Columns:

Column	Type / values	Meaning
`rally_number`	integer, 1-based, chronological ? no gaps, no repeats	rally index
`start_mmss`	`m:ss.mmm` (e.g. `1:12.40`)	rater-entered serve-contact time
`end_mmss`	`m:ss.mmm`	rater-entered ball/shuttle-dead time
`start_time`	decimal **seconds** (e.g. `72.40`) ? **AUTO-FILLED** from `start_mmss`	machine time
`end_time`	decimal **seconds** ? AUTO-FILLED from `end_mmss`	machine time
`shots_count`	integer	total racket/paddle/bat contacts (serve = shot 1)
`ending_reason`	`winner` | `forced_error` | `unforced_error` | `service_fault` | `let` | `other`	WHY it ended
`last_shot_outcome`	`in` | `net` | `out` | `n_a`	WHERE the last shot landed (`n_a` for service_fault/let)
`score_before`	optional, `"A-B"` ? blank if not confidently readable	score before
`score_after`	optional, `"A-B"`	score after
`notes`	optional free text ? **REQUIRED when `ending_reason = other`**	edge-case note

Times (mm:ss ? decimal seconds): raters type only `start_mmss` / `end_mmss`. The template Google Sheet fills `start_time` / `end_time` with this formula (shown for `start_time`, row 2):

6. user (2026-06-19T17:18:17.524Z)

Video Recording Guidelines (DRAFT ? thresholds pending empirical data)

Status: ? Draft / placeholder · **Date:** 2026-05-30
Why this exists: highlight quality is bounded by *input* quality. To get reliably good auto-generated highlights, the source video must be recorded a certain way (camera height, court position, resolution, frame rate, stability, framing). We want to (1) **derive** the minimum acceptable capture spec **empirically** from the precision/recall of the solutions we build, and (2) **publish user-facing best practices** so a user knows how to record footage that the product can index well.

> **This is a tracked thought, not a finished standard.** The exact numeric thresholds below are **placeholders** to be replaced with values measured against our golden set. See "How we'll set the real numbers" (§3). Captured now (per the team's working agreement) so it isn't

lost; to be revisited.
>
> **Update (2026-06-02):** the **frame-rate** and **camera-geometry** rows now carry **literature-backed starting values** (cited inline). They remain *starting points* to confirm empirically per §3. The golden-set **corpus** diversity spec (what footage to collect, not just how users should record) is now in GOLDEN_SET_PHASE1_VIDEOS.md.
>
> **Update (2026-06-07):** **AUDIO is now a first-class capture requirement** (new row below), per ADR-007 ? we will use the shuttle "thwack" as a rally cue (audio+visual fusion lifts recall). This is a hard "record with audio on, mic unobstructed" ask, not a placeholder; efficacy on real GoPro audio is being validated (experiment E1 in [AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md)).

?? THE #1 CAPTURE RULE: ONE MATCH PER FRAME (single court)

> **For the best auto-highlights, record a SINGLE match on a SINGLE court ? with no other matches
> playing in the background or on adjacent courts.** This is the happy path the product is tuned for,
> and as of **2026-06-18 it is MEASURED, not a guess:**
>
> - On a golden **multi-court** clip, the rally detector fired **87 "rallies" against 43 real ones ?
>   precision ? 0.20 (~80% false positives)**, almost all of them background-court games it mistook
>   for play.
> - On **single-court** clips, precision is markedly higher and rally boundaries are tight; local even
>   **out-recalls Gemini** on normal single-match play.
>
> **Multi-court is a KNOWN BOTTLENECK we are actively working to fix** (player-state + court-occupancy
> signals ? see RALLY_DETECTION_GAP_CLOSURE.md). Until that lands,
> **single-match capture is strongly recommended** ? a reel made from busy multi-court footage *will*
> include false "rallies" from the neighbouring games. **North star (for now): consistently beat Gemini
> on the single-match happy path.**

---

1. Why input capture matters (grounded in what we already know)

ADR-004 found a concrete, capture-dependent failure mode: on **broadcast** footage the camera *tracks the players*, so player pixel-motion dips during rallies and spikes on pans/replays ? making player-motion a poor rally proxy. The lesson generalizes: **the pipeline's accuracy is conditional on how the video was shot.** A **static, full-court** camera (typical of a fixed tripod recording) behaves very differently from broadcast ? and is likely much friendlier to both the motion/detector stages and the shuttle-tracking substrate (WASB/TrackNet, ADR-001).

So capture guidance isn't cosmetic ? it directly moves precision/recall.

---

2. The dimensions that likely matter (hypotheses to test)

Dimension	Why it plausibly affects P/R	Placeholder guidance (UNVERIFIED)
Resolution	Shuttle is tiny & fast; too few pixels ? detector misses it	? **1080p** preferred; **720p** likely floor; 2K/4K may help shuttle tracking but ? compute/cost
Frame rate	Motion blur; boundary precision; fast-ball detection	? **30 fps**

badminton/pickleball; **? 60 fps table tennis** ? 30 fps collapses fast-ball detection (**22.5% ? 93.2%** at 60 fps; 120 ? 60, [Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6)) |
Camera position	Static full-court view avoids the broadcast camera-tracking trap (ADR-004)	**Fixed tripod**, elevated, behind/above one baseline, whole court in frame
Camera height & angle	Occlusion of players/shuttle; court keystoning	**3?5 m** elevated (~4.5 m), **~30° down-tilt**, behind one baseline, whole court + net headroom ([badminton-CV setup](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/))
Framing / margins	Cropping the court or players breaks detection & relevance check	Entire court + a margin; players never leave frame
Stability	Handheld jitter inflates motion signals	Tripod / fixed mount; no panning or zoom during play
Lighting / exposure	Blur & contrast affect detection and the existing blur/relevance validators	Even indoor lighting; avoid backlight/glare; minimize motion blur
Single match in frame (no background games) ? ? MEASURED, the #1 rule	Adjacent/background-court games get detected as false rallies ? this is the **dominant precision failure**	**One match per frame, single court.** Multi-court precision measured at **?0.20** (vs ?0.31 single, tighter boundaries); multi-court fix in progress ? see RALLY_DETECTION_GAP_CLOSURE.md
Court contrast / background	Relevance check uses court color; busy crowds add false motion	Clear court markings; minimal moving background where possible
Codec / container	Pipeline already validates MP4 + decodes H.264/HEVC	MP4 (H.264/HEVC) as today; align with `archives/MONETIZATION_AUDIT.md` codec notes
Audio (REQUIRED ? ADR-007)	Shuttle "thwack" hits are a strong rally cue; audio+visual fusion lifts rally *recall* (the WASB bottleneck) ? see ADR-007 / archives/research/AUDIO_RALLY_DETECTION_PLAN.md	**Record with AUDIO ENABLED** (GoPro default ? do not mute). Keep the **mic unobstructed**; **avoid aggressive wind/noise reduction** that flattens the sharp impact transient. Camera reasonably close to the court helps SNR (shuttle hits are quiet).

These map onto checks that **already exist** in `backend/pipeline/validators/` (resolution, fps, blur, scene-cut, court relevance) ? so several can be enforced/measured programmatically rather than only advised.

---

7. user (2026-06-19T17:18:19.827Z)

Owned-model implementation plan (roadmap) ? DRAFT for owner review

Status: PLAN / roadmap. No owned-model *implementation* starts until the owner greenlights a specific
milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a training source).
Grounded in: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO + learned prototype
(`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy workflow (2026-06-09).

Design principle (owner directive): *envision now, build later*
Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be **architecturally envisioned
now** so we never hit a mammoth refactor. We **design the seams** (data contracts, export boundaries, the
event store) up front and **implement behind them** when each milestone is greenlit. Punting a *feature* is fine;
punting its *seam* is not.

North Star (reframed per PR #61 review)
- **Immediate deliverable:** a **training framework + harness** that produces a model with **very high precision/
  recall for extracting highlights + rallies from a clip**. *Quality is the first-order objective.*
- **Eventual product:** a **single, sport-agnostic, conversational** video model ? upload a clip, then ask

contextual questions ("longest rally", "make a clip of all service faults", "rallies over 20 shots").
- **On-device is LATER** (a model-compression goal so power users save upload cost/hassle) ? definitely on the roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
- **The moat = the consented dataset + the eval harness, not the weights.**

Hard guardrails (non-negotiable ? "never corrupted")
- **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT / Apache-2.0 / BSD only.** (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral), **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.
- **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public grounding-CoT datasets that were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision (a real, binding cost).
- **Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.**
- ? **Base-model pretraining provenance is unauditable for *every* open VLM (incl. Qwen3-VL).** Our "no paid-LLM training" rule is enforceable at *our fine-tuning data* layer, not the base's pretraining ? an **explicit owner risk-acceptance** is required to use any open VLM. (Governed by the base's license, not our pipeline.)

The model strategy (verified 2026-06-09)
Framework ? base (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge (runs SFT/GRPO, emits weights); a *base* is the metal you forge. You pick one framework and feed it any clean base.
- **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio + timestamp-grounding + GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core; *avoid its AGPL Studio UI*) as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-time snapshot ? re-verify at impl.
- **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-video context, purpose-built **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm the exact size's LICENSE pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but its Prohibited-Use posture is unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not the primary extractor. **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is cheap *only within timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-specific).
- **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict temporal boundaries** (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries"). **Boundary accuracy IS the product.** The head (`rally_seq_proto.py`) is already de-risked on our corpus (held-out AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params (trivially compressible later), and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So: **head now ? VLM later**, strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes, head refines boundaries) over wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*

54 ## Conversational video-QA ? build the bookkeeping NOW, punt the model
55 **Punt the conversational *feature*** until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI
56 over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn** (expensive; bad at exact
57 counting; the strong teachers are license-forbidden). **But build the *seam* now** (the brief requires it):
58 - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits structured timestamped
59 spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-SQL + ffmpeg clip-cut). The
60 cited queries ("longest rally", "all service faults", "rallies > 20 shots") are **structured-numeric ? SQL, not a
61 VLM job** (SQL counts exactly where VLMs hallucinate).
62 - **Designed-now seam:** make the **per-video event-index schema a first-class data contract** alongside the M0.1
63 corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count, ending_reason/fault_tags,
64 derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy (e.g. `service_fault`) up front**
65 even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration): `play_segments` today has
66 `video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights` isn't a table yet ? so it's an
67 *extension with new columns + a highlights table*, not a no-op. **Punt** the NL parser/agent until quality clears the bar.
68 - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our structured index (allowed ?
69 it reasons over our data, it is not trained on its outputs).
70

8. user (2026-06-19T17:18:19.890Z)

1 # Post-#166 Work Items, Milestones, Risk Assessment, and Sequencing Plan
2
3 **Date:** 2026-06-14 (post-merge of #166)
4 **Status:** Plan for owner review first. Execution only after explicit owner approval ("review first and then execute"; "I want a plan detail as to what are the next items planned, please execute only after I explicitly say so").
5 **Purpose:** Per owner directive after #166 merges: stop the prior babysit loop, deliver this as a new project MD PR for review (with full risk assessment like the T0-T5 in the archived audit, sequencing rationale, and broken into milestones), then (only after review + explicit go) execute one PR per slice off `master`.
6
7 **Key outcomes from #166 (merged c558372):** Hardening loop (T1-T5 + finale #165) COMPLETE and archived. NEXT_STEPS reframed with explicit P3 start signal (harness LOVO litmus for person features) + full prioritized candidate list. Gate A wired (opt-in, default-OFF safety per served regression litmus). Loop scheduler (019ec6d07da5) deleted; no replacement created. Handoff and archived audit updated terminal. Master also includes #167 (packaging).
8
9 **Sources (live, read these for details):**
10 - `docs/NEXT_STEPS.md` (prioritized 1-6 + rally track pick-up order)
11 - `docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3 learned fusion + P4 audio)
12 - `docs/EXPERIMENT_HARNESS.md` (A/B, shadow, LOVO-honest `compare`, promotion gate, flag default-OFF, rollback=config flip)
13 - `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (M0.1?M0.4 Phase-0, event-index contract, TAL head first, ms-swift + Qwen3-VL, guardrails)
14 - `docs/BACKLOG.md` (rename ?, cost benchmark as largest open risk, detector follow-ups, golden subproject, etc.)
15 - `docs/DEFERRED_HARDENING_PLAN.md` (remaining: train/eval separation guardrails)
16 - Archived: `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (T0-T5 risk style + Living Log precedent), `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md`, CLAUDE.md, `docs/HOW_RALLY_DETECTION_WORKS.md`
17
18 **Process guardrails (non-negotiable, carried forward):** ONE PR per slice/item off

`master`; no stacking. Branch, implement (flag-gated default-OFF for any quality/cue changes), run available tests + post-change guards (syntax/ast here; owner re-runs full `run_all_tests.py --cov...` + real-video + golden LOVO/ablation on hardware), commit descriptively, push, create PR, babysit (poll/fix/reply via MCP + handoff pattern) until owner merge observed ? then advance. Never self-merge. For quality experiments: use harness `compare` (LOVO + honest stats/CIs/sign test), `run_regression.py` (exit 0/1/3 gate), record to `eval_baselines/experiment_leaderboard.json`. Promote only on measured lift + no regression. Container limits: use `python -c "import ast; ..."` + note "All green where executable. Owner re-runs full on hardware."
19
20 ### Transition Record (from #166 merge + owner stop-loop directive)
21 #166 merged (c558372, LGTM 4702279497). Per explicit owner request after #166: "stop this loop and send a PR for these work items as a new project md file so we can review first and then execute this and do assess the risk so we can sequence them well. Would be good to break them into milestones." Scheduler 019ec6d07da5 deleted; no replacement. Handoff trimmed to pointer (per #165 review intent); detailed record here + in plan. All T1-T5 + #165 + #166 transition complete. Clean slate for review of this plan before any further execution.
22
23 ---
24
25 ## Work Items Catalog (with risk per T0-T5 precedent)
26
27 Risk levels (from archived audit §3 style):
28 - **LOW**: docs-only, config, pure additive tests, no behavior impact.
29 - **LOW-MEDIUM**: new/expanded positive tests or thin pure-logic glue.
30 - **MEDIUM**: new runtime paths (even if gated/flag default-OFF), new MIT/Apache deps, feature persistence, small harness increments, real-video validation.
31 - **MEDIUM-HIGH**: DB migrations + new data contracts (event store), train/eval guardrails (correctness of moat), compliance changes to training sources.
32 - **HIGH**: anything touching "paid-LLM as training data" guardrail, unquantified large costs (Gen-2 VRAM/$), production default flips without lift proof, licensing/C3 blockers, multisport generalization hypothesis.
33
34 1. **Grow the golden corpus (NEXT_STEPS #1 ? highest leverage)**
35 Priority: multi-court badminton (target distro + ship target), then ?3 matches per new sport (TT first; pickleball labels exist but ball detector blocker). Adult sessions only (minors-exclusion). Per video: label (Roora v8 + guidelines) ? curate import-local (Proprietary license) ? re-run harness.
36 **Risk: LOW** (mostly owner/data-collection + collector cross-repo; import is existing). Small-N statistics power improves with n=10+ (sign test reachable).
37 **Verification:** manifest growth, re-harness LOVO/sign test on held-out folds, owner real footage.
38
39 2. **Ratify ship-gate target (NEXT_STEPS #2 + BACKLOG + OWNED M0.4)**
40 Floor = beat heuristic LOVO ~0.480 (necessary). Target informed by data: multi-court wrapper 0.66 (the number that makes owned worth serving); clean 0.93 is ceiling, not target. Shape suggestion: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic".
41 **Risk: LOW** (docs/decision only).
42
43 3. **Rally quality P3 (learned person fusion) + P4 audio (FUSION §10 + EXPERIMENT_HARNESS P2/P3)**
44 P3: integrate person features (from the P2 FusionSegmenter + `fusion_features`) into the harness `compare` for LOVO litmus on golden videos; report ?F1 + honest stats/CIs/sign test. Eager/windowed compute is in place from P2. If lift: optional served gating (flag default-OFF).
45 **Investigation note (updated per latest review):** The LOVO compare driver (`fusion_compare.py`), golden extractor (`fusion_golden.py`), ablation runner (`ablation.py`), and related tests (`test_fusion_compare.py` etc.) **are present** in the tree (verified via `git ls-files` + ls on backend/eval/ and tests/). Person feature computation lives in `backend/pipeline/segmenters/fusion_hybrid.py::_person_features` (and `backend/pipeline/detectors/fusion_features.py`). The remaining in-container P3 work for M0 is wiring the person features into the `compare` path (leveraging the existing harness seam for extra feature columns in the LogisticRegression) + running the litmus on the golden substrate. GPU extraction over the full corpus (to produce the real feature JSONs) remains the hardware-gated step. This makes the M0 placement of item 3 even more appropriate as a quick in-container

win (the measurement machinery is shipped; the integration + run is the task).
46 P4: wire audio (`hit_detector`) + re-eval (prior E1 found weak; cheap re-probe). Full learned promotion only on measured lift.
47 **Risk: LOW-MEDIUM for pure in-container harness litmus glue + test; MEDIUM overall** (including gated served path + owner real LOVO).
48 **Verification:** harness `compare` (deterministic on synthetic + golden), no-regression invariant, owner real-video + ablation; `run_regression.py` gate.
49
50 4. **M0.4 Gen-0 TAL harness (OWNED M0.4 + NEXT_STEPS #3)**
51 Swap Gen-0 LogReg for ActionFormer (or ASFormer; MIT, lightweight, minutes on 2070S) inside existing harness (`training/gen0/harness.py` reuse `metrics.match_intervals`). Per-video threshold calibration (LOVO-vs-oracle gap visible per fold). One-command train?eval?ship-gate. Define ship target up-front (see item 2).
52 **Risk: MEDIUM** (new MIT dep in training, code change to harness, but local $0, reuses featurizer/metrics, no production path until gate passed).
53
54 5. **Owned Phase-0 foundation ? M0.1 corpus spine + event-index contract (OWNED M0.1)**
55 Canonical JSONL per rally (video_id MD5, rally_ordinal, sport, start/end, shot_count, ending_reason/fault_tags, label_type, source, consent/training_opt_in, split=BY MATCH, trajectory_ref, ...). Wire to durable per-video **event store** (the conversational-QA seam: "extract once, query many"). DB delta: extend `play_segments`, new `highlights`/events table + `PRAGMA user_version` migration. Reserve fault/event taxonomy now (service_fault etc.). No Gemini-sourced rows in training views.
56 **Risk: MEDIUM-HIGH** (real DB migration + new first-class data contract affecting all future conversational features; must be correct early).
57
58 6a. **M0.3 Gemini-silver purge from training paths (OWNED M0.3)** ? retarget `distill_local`/`calibrate_local` etc. to human + open-teacher labels only (Gemini eval/fallback forever). Core moat-correctness code change.
59 **Risk: HIGH** (guardrail "never train on paid-LLM").
60
61 6b. **Train/eval separation guardrails ? pure-logic core (DEFERRED #13 + BACKLOG)** ? `backend/eval/splits.py::assert_disjoint(train_ids, eval_ids)` + unit test + raise-on-overlap. Split unit = match. (Defer collector manifest `split` field wiring to later cross-repo step.) In-container, zero behavior risk.
62 **Risk: LOW-MEDIUM** (pure logic + tests; unblocks clean baselines).
63
64 6c. **WASB C3 + related commercial blockers (tracking only)** ? first-class tracked item (own-weights or clean detector; multiplies per sport). No code change in this repo until resolved upstream.
65 **Risk: (commercial blocker, not code risk).**
66
67 7. **Multisport thread + clean detectors (NEXT_STEPS #4 + OWNED + BACKLOG)**
68 Ingest Roora pickleball/TT CSVs (`import-local --normalize`). Merged-prototype LOVO across badminton+TT once TT has ?3 matches. Identify clean MIT/Apache/BSD ball/shuttle detector per sport (WASB C3 is unresolved blocker for some paths; TT-on-WASB serving gated). Single-model-multisport-temporal-generalization is unproven hypothesis (validate on 2nd-sport corpus).
69 **Risk: MEDIUM** (data + per-sport detector hunt; C3 is commercial blocker).
70

9. assistant (2026-06-19T17:18:22.737Z)

Now let me gather cross-reference evidence: check for already-archived copies and inbound citations across the repo.

10. user (2026-06-19T17:18:23.402Z)

docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md

```
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py
docs\archives\quality-iterations\README.md
```

11. user (2026-06-19T17:18:25.121Z)

```
CLAUDE.md:10:- **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
(+ the
CLAUDE.md:20:  eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
CLAUDE.md:33:  greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
README.md:275:[Omitted long matching line]
docs\COMMERCIALIZATION.md:43:[Omitted long matching line]
docs\COMMERCIALIZATION.md:47:| C11 | **Input-quality enforcement** | **P2** | ? | Per-video
observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). |
Turn more `VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
docs\COMMERCIALIZATION.md:159:  narrows the addressable market ? enforce/guide it (C11,
`VIDEO_RECORDING_GUIDELINES`).
docs\BACKLOG.md:12:## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
`NEXT_STEPS.md`)
docs\BACKLOG.md:52:*(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
docs\EVALUATION.md:21:> bottleneck. See `docs/TRACKNET_WSL_SETUP.md`.
docs\DESKTOP_APP_PLAN.md:6:> **Relation to existing docs:** **peer-of**
[`PLATFORM_ARCHITECTURE.md`](PLATFORM_ARCHITECTURE.md) (this is the **`LocalDesktop`
ComputeBackend** made into a product) · **realizes the L0 tier** of
[`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) · **consumes**
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md) (the smaller/export-
clean model is one answer to the compute crux) · bound by
[`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) licensing guardrails.
docs\DESKTOP_APP_PLAN.md:14:[Omitted long matching line]
docs\DESKTOP_APP_PLAN.md:48:[Omitted long matching line]
docs\DESKTOP_APP_PLAN.md:119:- **If CPU-only is too slow on a typical laptop, the simple "wrap
the pipeline" path does not exist** ? it becomes "ship a smaller owned model" (a much larger
effort, ties `OWNED_MODEL_IMPLEMENTATION_PLAN`) or "GPU-tier only" (shrinks the addressable
market). The doc names this honestly rather than assuming the wrap is free.
docs\DESKTOP_APP_PLAN.md:138:2. **(gates P1) If CPU is too slow, which fork?** tiered GPU-
fast/CPU-slow UX · invest in a smaller exported owned model (`OWNED_MODEL_IMPLEMENTATION_PLAN`)
· GPU-tier-only product (narrower market).
docs\DESKTOP_APP_PLAN.md:149:- [ ] **P0** produced a measured throughput table (CPU / MPS /
CUDA-native) and an owner-accepted go/no-go verdict (incl. whether a smaller model is required).
*(P0 alone is a valid stopping point if the verdict is "not feasible without the owned model" ?
then this doc hands off to `OWNED_MODEL_IMPLEMENTATION_PLAN`.)*
docs\DESKTOP_APP_PLAN.md:157:[Omitted long matching line]
docs\GOLDEN_DATA_SHARING.md:52:[Omitted long matching line]
artifacts\rally_tal_gen0\0.1.0\manifest.json:186:      "ship-gate F1@tIoU target UNDEFINED
(OWNED_MODEL_IMPLEMENTATION_PLAN M0.4) \u2014 beating the floor is necessary, not sufficient",
docs\GOLDEN_SET_VENDORING.md:8:[Omitted long matching line]
docs\GOLDEN_SET_VENDORING.md:34:> **Honest tradeoff:** a licensed/owned corpus may not perfectly
mirror real users' cameras, courts, and lighting. A green regression on our corpus is therefore
*strong evidence*, not a *guarantee*, for production traffic. Revisit if field behavior diverges
```

```
from corpus behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by
steering users toward capture conditions our corpus represents.
docs\GOLDEN_SET_VENDORING.md:188:2. **Corpus representativeness** ? how well does the
owned/licensed collection mirror eventual user capture? (Ties directly to
VIDEO_RECORDING_GUIDELINES.md.)
docs\HARDENING_LOOP_HANDOFF.md:13:**#166 merged** (c558372). Transition record + P3 signal +
work-items plan (with risk assessment, milestones, sequencing for review-first execution) in
`docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md`. Loop stopped per owner directive
(no further schedulers). See that plan MD for full details and "How to use this plan".
docs\GOLDEN_SET_PHASE1_VIDEOS.md:4:**Companion docs:**
[GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (vendor process, ADR-006) ·
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) (capture spec the user-facing
guide derives from) · scoring in [EVALUATION.md](EVALUATION.md).
docs\GOLDEN_SET_PHASE1_VIDEOS.md:48:> These are **literature-backed starting points**; the
published minimums get finalized empirically against our golden set
(VIDEO_RECORDING_GUIDELINES.md §3).
docs\GOLDEN_SET_PHASE1_VIDEOS.md:96:Outdoor courts, phone-handheld/jitter, extreme angles, 4K,
unusual net/court variants, crowd-heavy tournaments. Add these once round-1 precision/recall
shows the weakest axis, then expand toward it (find the quality "knee" ?
VIDEO_RECORDING_GUIDELINES.md §3).
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:12:## Update (2026-06-09, post-
PR-#61-review) ? authoritative roadmap is now `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
docs\NEXT_STEPS.md:12:`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology).
**Latest blow-by-blow:** memory
docs\NEXT_STEPS.md:61:  `docs/ROORA_LABELING_GUIDELINES.md` (v8) ·
`docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
docs\NEXT_STEPS.md:85:   `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training
pool** (consent guardrail). Per video:
docs\NEXT_STEPS.md:115:- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:30:
`docs/VIDEO_RECORDING_GUIDELINES.md`).
docs\MULTI_SIGNAL_FUSION_PLAN.md:13:> and the labeling spec in
[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md).
docs\MULTI_SIGNAL_FUSION_PLAN.md:185:[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md
)), so held-shuttle moments are negatives
docs\MULTI_SIGNAL_FUSION_PLAN.md:262:- **P0 ? this design doc** +
[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md), indexed in
docs\PLATFORM_ARCHITECTURE.md:6:> ? **Superseded where it disagrees with
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
docs\PERSONALIZATION_PLAN.md:8:[OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_P
LAN.md),
docs\PMF_MARKET_RESEARCH.md:197:   `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:13:-
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (M0.1?M0.4 Phase-0, event-index contract, TAL head
first, ms-swift + Qwen3-VL, guardrails)
docs\README.md:1:- [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8)
vendor labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-
driven court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
docs\README.md:2:[Omitted long matching line]
docs\README.md:15:[Omitted long matching line]
docs\README.md:18:- [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
docs\ROADMAP.md:11:> [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
? a learned **per-frame TAL head**
docs\ROADMAP.md:122:> that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
docs\RALLY_QUALITY_RESEARCH.md:24:> [`EVALUATION.md`](EVALUATION.md),
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
docs\RALLY_QUALITY_RESEARCH.md:213:
[`ROORA_LABELING_GUIDELINES.md`](ROORA_LABELING_GUIDELINES.md)) ? court calibration, IAA, the
docs\RALLY_QUALITY_RESEARCH.md:473:  not grafted (§5.5 refutation)). Per
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md):
docs\RALLY_QUALITY_RESEARCH.md:565:> Tier 3 spins out into the owned-model milestone
([`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).
docs\RALLY_QUALITY_RESEARCH.md:579:
```

[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
docs\RALLY_QUALITY_RESEARCH.md:584: [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) · [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:116:`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (Gen-0 first), gated on n?5 + owner sign-off.
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:253: + `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`, after scaffolding).
docs\STORAGE_SHARING_MODEL.md:99:   more users ? more corrections. The owned-model roadmap (OWNED_MODEL_IMPLEMENTATION_PLAN) IS
docs\UI_PROPOSAL.md:4:**Date:** 2026-06-10 · **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION, PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES, reporting/spec.yaml, current API + static UI.
docs\UI_PROPOSAL.md:44:| A2 | **Upload & Process** | Drag-drop; inline recording-guidelines checklist (pre-upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing | B2 upload endpoint ? B1 job |
backend\reporting\spec.yaml:91: # Audio stream missing or disabled. Per VIDEO_RECORDING_GUIDELINES (and ADR-007)
backend\reporting\spec.yaml:104:   transient. See VIDEO_RECORDING_GUIDELINES.md (the "Audio (REQUIRED)" row) and
backend\reporting\spec.yaml:106:   faq_ref: "VIDEO_RECORDING_GUIDELINES"
backend\reporting\spec.yaml:110:   re-process after enabling audio (see VIDEO_RECORDING_GUIDELINES.md).
docs\archives\COMMERCIAL_READINESS_REVIEW.md:47:- Input Quality UX extended inside the existing per-video reports (audio-present signal from ffprobe + actionable `audio_disabled_or_missing` rule + guidance pointers to VIDEO_RECORDING_GUIDELINES) ? PR #50, merged.
docs\archives\COMMERCIAL_READINESS_REVIEW.md:82:   - Turn more VIDEO_RECORDING_GUIDELINES dimensions into stronger validators or report rules/footguns (stability via scene cuts? lighting via blur/relevance? camera geometry heuristics?).
docs\archives\decisions\DECISIONS.md:281:- New docs: `docs/GOLDEN_SET_VENDORING.md` (the full process v1) and `docs/VIDEO_RECORDING_GUIDELINES.md` (a tracked draft: capture-quality thresholds to be derived empirically from precision/recall, since input quality bounds output quality ? cf. ADR-004's broadcast-camera finding).
docs\archives\decisions\DECISIONS.md:325:5. **Make "audio always enabled" a capture requirement** (`docs/VIDEO_RECORDING_GUIDELINES.md`), since the cue
backend\pipeline\detectors\base.py:66:- docs/TRACKNET_WSL_SETUP.md and CODE_MAP.md (2.2 Detectors section)
tests\test_reporting_spec_faq.py:52:   elif file_part in ("VIDEO_RECORDING_GUIDELINES", "VIDEO_RECORDING_GUIDELINES.md",
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:38:8. Input Quality UX ? reframed as extend existing reports (audio readiness + capture guidance from VIDEO_RECORDING_GUIDELINES). ? (feat/extend-reports-input-quality + PR #50, merged)
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:188: - 2. Input Quality UX via reports extension (audio readiness signal + capture guidance surfaced in reports from VIDEO_RECORDING_GUIDELINES) ? #50

12. assistant (2026-06-19T17:18:30.250Z)

Strong signal. None of the five docs have an archived copy with their stem. All five are heavily cited as live/authoritative sources. Let me verify the two "plan" docs more closely for terminal status, and check the POST_166 doc's citation context plus NEXT_STEPS for whether it's superseded.

13. user (2026-06-19T17:18:31.812Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HARDENING_LOOP_HANDOFF.md-12-
docs\HARDENING_LOOP_HANDOFF.md:13:**#166 merged** (c558372). Transition record + P3 signal + work-items plan (with risk assessment, milestones, sequencing for review-first execution) in `docs/POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md`. Loop stopped per owner directive (no further schedulers). See that plan MD for full details and "How to use this plan".
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HARDENING_LOOP_HANDOFF.md-14-

14. user (2026-06-19T17:18:32.737Z)

8. **Housekeeping ? rename repo off "badminton" (BACKLOG ? + OWNED)**
   Owner-emphatic "ASAP" (project is multisport). Touches remote name, README/CLAUDE framing, any hardcoded, collector cross-refs.
   Risk: LOW-MEDIUM (docs + references + coordination; breaking for external references until done).

9. **PMF validation (NEXT_STEPS #5)**
   C13 academy interviews (add the two questions on multi-court + pay-per-match). Camera pilot at 2-3 warmest academies (consent at capture; demo via Gemini or human-polish reel). Field kit complete (flyer + playbook + answer sheet + pre-registered G/A/R signals + PROCEED/PIVOT/FALSIFIED thresholds). Owner-led (hired associate or direct).
   Risk: LOW (mostly non-code / owner field work; cheap vs engineering month).

10a. **In-container chores (low-regret, executable now)** ? WSL/Gemini remote-file GC + retention policy note; tests reorganization (per owner-specified grouping metric from prior reviews); other small docs/code hygiene from BACKLOG.
    Risk: LOW.

10b. **Hardware / open-risk items (triggered, owner-gated)** ? Quantify cost-to-Gen-2 (the single largest open risk: real VRAM/token/$ on A100/L40S before any owned serving); ByteTrack deprecation + real-footage person detector validation; golden subproject follow-ups that require hardware/corpus.
    Risk: MEDIUM-HIGH (unquantified cost is load-bearing; real video needed).

Risk Assessment Summary Table (unbundled per review)

ID	Item	Risk	Rationale / Deps	Verification
1	Grow golden corpus	LOW	Data effort + existing curate; consent guard	Manifest + LOVO/sign test improvement
2	Ship-gate target	LOW	Docs/decision; unblocks harness error	Explicit ratified number in plan
3	P3/P4 fusion litmus + harness (in-container core + gated)	LOW-MED (core) / MEDIUM (full)	General harness present; specific person-feature LOVO driver + GPU extraction are the work. Gated promotion.	harness `compare` + honest stats + regression gate + owner ablation/real LOVO
4	M0.4 ActionFormer harness	MEDIUM	MIT dep + harness change (local $0)	One-command + per-fold cal + ship-gate
5	M0.1 event-index contract + spine	MEDIUM-HIGH	DB migration + new contract	Round-trip, split-by-match, no Gemini train rows
6a	Gemini-silver purge (training paths)	HIGH	Core moat guardrail correctness	Training entrypoints retargeted; no Gemini-silver in train views
6b	Train/eval guard ? pure logic core	LOW-MEDIUM	`splits.py` assert + test (in-container)	Unit test + raise-on-overlap; defers collector wiring
6c	C3 / commercial blockers	(tracking)	Multiplies per sport; gates ship	Tracked in BACKLOG/OWNED; no code until resolved
7	Multisport + detectors	MEDIUM	Data + per-sport detector hunt (C3 multiplies)	?3 TT matches + cross-LOVO prototype
8	Rename	LOW-MEDIUM	References + remote + framing	Clean grep + owner remote action
9	PMF field	LOW	Owner field work (kit ready)	Interviews logged + signals thresholds
10a	In-container chores (GC, tests-reorg, etc.)	LOW	Small hygiene / docs	Grep-clean, tests re-org per owner metric
10b	Cost-to-Gen-2 benchmark + real validations	MEDIUM-HIGH	Unquantified $ (largest open risk); needs A100/L40S + real footage	Benchmark numbers + reports before any Gen-2 work

Overall sequencing risk mitigation (revised per course-correction): Front-load M0 in-container quick wins (high-value engineering with zero external blockers). Owner-led work (corpus growth as #1 leverage, PMF, rename) runs in parallel. Heavy/gated work (DB migration, purges, cost benchmark) only after corpus + ratified ship target + quick wins are in place. All

quality items flag default-OFF, promote only on measured lift via harness + regression gate + owner real-video. Owner explicit "proceed with X" required before any implementation.
106
107   ---
108
109   ## Milestones (revised per review course-correction ? front-load in-container quick wins)
110
111   **Milestone 0: Quick Wins (in-container engineering; unblocked by corpus, hardware, or additional owner gates ? do these now)**
112   These are the low-hanging, high-value items that can be executed in the dev container with the existing harness + pure logic. Per the rebuttal: front-load these so the cheapest engineering wins aren't buried behind heavy owned-model / corpus work.
113
114   1. **Ratify the ship-gate target (item 2).** Pure decision/docs. Clears the live hard-error in `training/gen0/harness.py` on "undefined F1@tIoU target". Numbers already exist in the data (floor ~0.480 heuristic, multi-court reference 0.66, clean ceiling 0.93). Highest-value quick win; also unblocks M0.4 gate and regression baselines.
115   **Risk: LOW**. In-container.
116
117   2. **Pure-logic core of train/eval guard (item 6b).** Implement `backend/eval/splits.py::assert_disjoint(train_ids, eval_ids)` + raise-on-overlap unit test (split by match). The spec is in BACKLOG/DEFERRED. Fully testable here; defer collector manifest wiring.
118   **Risk: LOW-MEDIUM**. In-container.
119
120   3. **Run / land the P3 fusion LOVO litmus (item 3 core).** Integrate person features into harness `compare` (or dedicated driver) and execute on existing/synthetic features + the 6 golden (LOVO + honest stats). Note: general experiment harness + P2 fusion segmenter/person persistence are present; the specific P3 litmus driver over fusion features is the implementation task (see investigation in catalog item 3). GPU extraction over full corpus is the hardware-gated part.
121   **Risk: LOW-MEDIUM** (in-container glue + tests).
122
123   4. **In-container chores from item 10a.** WSL/Gemini GC + retention policy; tests re-org (owner grouping metric); other small hygiene. Separate from hardware cost benchmark.
124   **Risk: LOW**.
125
126   *Parallel / owner-led (not primary agent engineering focus):* Grow the golden corpus (your #1 leverage item), PMF field visits (using the existing kit), repo rename. These run alongside M0 without blocking the quick engineering wins.
127
128   **Milestone 1 ? Rally Quality P3 completion + ActionFormer (after corpus growth + ratified ship target)**
129   - Complete P3 (GPU litmus on real corpus + any served wiring on proven lift) + P4 audio re-probe.

15. assistant (2026-06-19T17:18:43.132Z)

I now have enough evidence. Let me synthesize the verdicts.

Key findings:
- **POST_166** is a "Plan for owner review first... Execution only after explicit owner approval." It is a milestone/sequencing plan whose work items (M0.1-M0.4, P3/P4, corpus growth, rename) are still in-progress/pending. It is cited as a LIVE source by `docs/HARDENING_LOOP_HANDOFF.md`. This is owner-gated FUTURE work, not terminal ? KEEP_LIVE. The "#166 hardening loop COMPLETE" is just its origin context, not the doc's own status.
- **OWNED_MODEL_IMPLEMENTATION_PLAN** is explicitly the authoritative roadmap, cited in CLAUDE.md as authoritative, referenced everywhere. Status = PLAN/roadmap, no implementation started ? LIVE design ? KEEP_LIVE.
- **ROORA_LABELING_GUIDELINES** v8 ? "single label to use with Roora going forward," sole canonical copy, cited live in CLAUDE.md, NEXT_STEPS, README, MULTI_SIGNAL_FUSION_PLAN, RALLY_QUALITY_RESEARCH ? KEEP_LIVE (active reference).
- **TRACKNET_WSL_SETUP** ? operational setup/usage reference, cited live in EVALUATION.md, README, CODE_MAP, backend/pipeline/detectors/base.py ? KEEP_LIVE (reference).

- **VIDEO_RECORDING_GUIDELINES** ? draft but actively tracked, MEASURED #1 rule updated 2026-06-18, cited live across COMMERCIALIZATION, BACKLOG, ROADMAP, PMF, UI_PROPOSAL, backend/reporting/spec.yaml, a test ? KEEP_LIVE (in-progress tracked thought).

None have an archived copy with their stem; none are full duplicates of a COMPLETED archive.

16. user (2026-06-19T17:19:12.058Z)

Structured output provided successfully

17. assistant (2026-06-19T17:19:25.246Z)

All five docs in my batch are KEEP_LIVE. Summary of findings:

None of the five docs are in a terminal (DONE/REJECTED/superseded) state, none has an already-archived copy under its stem, and all are cited as live/authoritative sources ? so none should move to `docs/archives/`.

- **ROORA_LABELING_GUIDELINES.md** ? Active canonical v8 vendor spec ("single label to use with Roora going forward", "sole canonical copy"), with open `TODO(avidullu)` owner items. Cited live in CLAUDE.md, NEXT_STEPS, README, MULTI_SIGNAL_FUSION_PLAN, RALLY_QUALITY_RESEARCH.

- **TRACKNET_WSL_SETUP.md** ? Operational setup/usage reference for the live WSL2 shuttle-detector substrate; pointed to from `backend/pipeline/detectors/base.py` and EVALUATION.md. Reference doc ? KEEP_LIVE.

- **VIDEO_RECORDING_GUIDELINES.md** ? Self-described "tracked thought, not a finished standard" with placeholder thresholds; actively updated (MEASURED multi-court rule 2026-06-18, ADR-007 audio row) and wired into `backend/reporting/spec.yaml` + a passing test (`tests/test_reporting_spec_faq.py`). In-progress ? KEEP_LIVE.

- **OWNED_MODEL_IMPLEMENTATION_PLAN.md** ? Status: PLAN/roadmap, no implementation started; CLAUDE.md names it the authoritative owned-model roadmap that supersedes older docs. The archived `OWNED_MODEL_TRAINING_STUDY.md` defers TO this doc, not vice-versa. Owner-gated not-yet-started design = LIVE.

- **POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md** ? Note the trap: it records that the #166 hardening loop is COMPLETE, but its OWN status is a forward-looking "plan for owner review first; execution only after explicit owner approval" of unexecuted milestones (M0 quick wins, M0.1-M0.4, corpus growth, rename, PMF). `docs/HARDENING_LOOP_HANDOFF.md` cites it as the live work-items plan. Owner-gated pending plan ? KEEP_LIVE, not terminal.

Relevant absolute paths:
- C:\Users\avidu\Projects\badminton-highlight-indexer\docs\ROORA_LABELING_GUIDELINES.md
- C:\Users\avidu\Projects\badminton-highlight-indexer\docs\TRACKNET_WSL_SETUP.md
- C:\Users\avidu\Projects\badminton-highlight-indexer\docs\VIDEO_RECORDING_GUIDELINES.md
- C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
- C:\Users\avidu\Projects\badminton-highlight-indexer\docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md